

DEVELOPMENT SET

97 intent groups, 130 questions, every candidate graded. The ladder, cheapest first.

System	What it is	nDCG@10	encode corpus
BM25	the store's word-matching score	0.396	no model
BM25F	BM25 over weighted fields	0.471	no model
potion-base-32M	static model, one vector per word	0.471	–
potion-retrieval-32M	static model, tuned for search	0.479	1.5 s
all-MiniLM-L6-v2	small transformer	0.494	9 s
Qwen3-Embedding-0.6B	large dense model	0.629	1,556 s
mDenseOn	multilingual dense model	0.630	569 s
DenseOn	dense retrieval model	0.640	47 s
LateOn	late interaction, per-word scores	0.695	–

SEALED HELD-OUT SET

62 intent groups written before any model was compared, opened once. Natural-language questions.

System	What it is	nDCG@10
DenseOn	dense retrieval model	0.689
BM25F	BM25 over weighted fields	0.371
potion-retrieval-32M	static model, tuned for search	0.353

THE SHIPPED CONFIGURATION

One deterministic run: exact cosine scan over all 5,301 vectors, no approximate index.

System	What it is	nDCG@10
DenseOn alone	scan, top 10	0.634
DenseOn + LateOn	scan for 30, rerank 30 in 0.6 ms	0.685